# Vision Transformer-based Classification for Lung and Colon Cancer using Histopathology Images

*Abstract*—**In this day and age, a considerable number of fatalities are caused by colon and lung cancer. Although their appearance simultaneously is rare, the likelihood of cancer cells spreading between these two organs in the absence of early diagnosis is relatively significant. The second highest cause of death worldwide is cancer. Colon and lung cancers are the most typical and lethal malignancies among the many different forms. Historically, physicians had to go through a lengthy and time-consuming process to evaluate histopathological images and identify cancer cases; however, current technology options make it possible to complete this process more quickly. This study classified the histopathological imagery LC25000 gathered at the Tampa, Florida-based James A. Haley Veterans Hospital of Lung and Colon Cancers using several Convolutional Neural Network (CNN) variants as well as the more recent vision transformer architecture. Based on the outcomes of this study, it has been observed that using the Vision Transformer provides exceptional performance outperforming deep learning CNN, VGG19, and Resnet 50 algorithms with 100% accuracy.**

*Index Terms*—**Histopathology, Vision Transformer, CNN, VGG19, ResNet, Cancer**

## I. Introduction

CANCER is a group of disorders brought on by unchecked cell division as a result of frequent gene alterations. Several factors, including behavioral traits like high body mass index, tobacco and alcohol use, exposure to physical carcinogens like ultraviolet rays and radiation, as well as specific biological and genetic carcinogens [1] like Human Papilloma Virus (HPV), are to blame for these genetic alterations. Nearly 10 million people died from several types of cancer worldwide in 2020 [2]. Lung and colon cancers are the most common cancer among them [2]. They continue to be the top cause of death worldwide, accounting for almost 2.8 million deaths in 2020 [1].

According to previous reports, patients with lung cancer develop colon cancer most frequently [3], [4]. A series of steps are involved in the treatment of cancer, including early detection, screening, therapy, palliative care, and survivorship care. A cancer diagnosis made before the disease has spread to other organs can considerably lower the death rate and improve treatment options. Prior to World Cancer Day, the World Health Organization (WHO) updated its recommendations, emphasizing the need of early diagnosis for effective cancer therapy (1). However, the majority of the population in low- and middle-income countries lacks access to adequate diagnostic facilities. Advanced computational models and AI tools have become a great solution to this. Machine learning-based image identification technology can increase the accuracy of diagnosing incurable diseases, giving doctors more crucial time to save patients while also providing solid data to back up their findings. It can also lessen the workload for medical staff even more. Computational approaches and image analysis techniques have been used previously in several disease diagnoses including- infectious disease [5], neurodegenerative diseases [6], cancers. In this study, we have attempted to use the state-of-the-art transformer models to further enhance prediction power of these AI algorithms thereby reducing treatment response time. The rest of the paper is organized as follows. Section II discusses the background associated with these diseases and deep learning models used to diagnose diseases, Section III explains the dataset used in this research, Section IV highlights the algorithms used in the experiments, Section V discusses experiments and presents the result analysis, and Section VI presents conclusion and future work.

## II. Literature Review

Two of the most prevalent cancers in the world are lung and colon cancer, and early diagnosis is essential to enhancing patient outcomes. The conventional procedure for detecting cancer is a histological examination of tissue samples, and developments in digital pathology have made it possible to apply computer-aided techniques for the automated processing of histopathology images. Image classification is typically done in feature maps, building high-performance image classification algorithms depends heavily on effective feature extraction. According to the features they employed [7], we may usually classify existing image classification methods into three primary categories. (i) Handmade feature-based techniques that require human-engineering skill and expert knowledge, (ii) Approaches based on unsupervised feature learning, for example, principal component analysis, k-mean clustering algorithm, and so on, (iii) Deep learning methods, like CNN, AlexNet, VGGNet, ResNet, etc. When it comes to image classification, deep learning algorithms outperform other feature extraction methods [8]–[10]. For the most part, current state-of-the-art methods rely on supervised learning to provide high-quality feature representations. Deep learning has become a very popular approach for visual scene classification due to advancements in artificial neural networks, particularly deep convolutional neural networks [11]. Deep learning features, as opposed to conventional handcrafted features, are automatically learned from data using a general-purpose learning procedure via deep-architecture neural networks [12]. This is in contrast to traditional handcrafted features, which demand a significant amount of engineering skill and domain knowledge [13]. On the other hand, deep learning models with many processing layers can learn more potent feature representations of data with multiple degrees of abstraction than unsupervised feature learning techniques, which are often shallow-structured models. This is the key advantage of deep learning methods. Additionally, deep learning techniques have proven to be particularly effective at unearthing complex structures [14] and discriminative information concealed in high-dimensional data, and features.

In recent years, deep learning methods have shown promising results in several field [15]–[17] including in the automated detection of lung and colon cancer from histopathological images. Kumar et al (2022) proposed a handcrafted and dense feature extraction techniques with deep learning methods. Their suggested random forest classifier can recognize lung and colon cancer tissue with accuracy and recall of 98.60%, a precision of 98.63%, an F1 score of 0.985, and ROC-AUC of 0.1 using deep features taken from DenseNet-121 and outperformed other state-of-the-art methods [18]. In another study, Masud M. et al. proposed modern Deep Learning (DL) and Digital Image Processing (DIP) techniques. The 2D-DFT and 2D-DWT methods are used to extract features, which are then combined with a CNN classifier in the model. The obtained findings demonstrate a maximum accuracy of 96.33% for the proposed methodology to identify cancer tissues [9]. In 2020, A lung cancer diagnostic technique based on nodule region of interest (ROI)-based feature learning with CNN was described by Suresh and Mohan. They gathered CT scan pictures from the databases of the Infectious Disease Research Institute (IDRI) and the Lung Image Database Consortium (LIDC), and they used Generative Adversarial Networks (GANs) to create more images to expand the sample. Using their proposed model, they were able to attain a maximum classification accuracy of 93.9% [19]. Wang et al. (2019) proposed a hybrid deep learning approach for lung cancer detection that combines the advantages of CNN and recurrent neural networks (RNN). The proposed approach achieved an accuracy of 89.8% for lung cancer detection [20].

The science of computer vision has undergone a revolution with the development of Vision Transformer (ViT), a potent model for picture classification tasks. The transformer architecture originally created for natural language processing is used by ViT to process image patches, as described by Dosovitskiy et al. in "An Image Is Worth 16x16 Words: Transformers for Image Recognition at Scale" [21]. ViT learns intricate patterns inside images by using self-attention processes to record both local and global visual aspects. The model has produced competitive results when compared to conventional convolutional neural networks (CNNs) and has displayed amazing performance on a variety of picture categorization benchmarks. To further advance image classification tasks, researchers have investigated various ViT setups, such as pre-trained models on massive datasets, transfer learning techniques, and hybrid architectures. For example, Jiangyun Li et al. (2022) introduced the Vision Transformer Enhanced (TransBTSV2) model, which performed better than conventional convolutional neural networks (CNNs) in the classification of medical images for the segmentation of brain tumors, liver tumors as well as kidney tumors [22].

## III. Dataset Description

The Lung and Colon Cancer Histopathological Image Dataset also known as the LC25000 dataset is a useful tool for cancer researchers and clinicians. Images for the LC25000 collection were gathered at the Tampa, Florida-based James A. Haley Veterans Hospital. Later, it was assembled by Borkowski et al. in 2019 [23]. This collection includes high-resolution digital photographs of lung and colon cancer tissues that have been meticulously annotated by board-certified pathologists with regard to their pathological characteristics. The dataset is publicly available to download [24]. It has five classes and 25,000 histopathology images. The images, which were originally $1024 \times 768$ pixels but were squared up to $768 \times 768$ pixels, are in the JPEG file type. The initial example dataset consists of 750 total images of lung tissue with 3 types and 500 total images of colon tissue with 2 types and is enhanced to 25,000 using the Augmentor program. The dataset has great potential to support the development of new diagnostic and therapeutic approaches for lung and colon cancer, as well as to improve our understanding of cancer biology. The detailed characterization of the tumor shape and the identification of important aspects linked to disease development and treatment response are made possible by the accurate annotation of the images, which is a fundamental strength of this collection. Sample images of lung cancer are shown in Figure 1 while colon cancer are shown in Figure 2

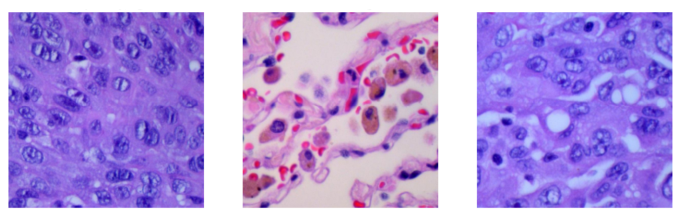

Fig. 1: Sample Histopathology images of three lung cancer types

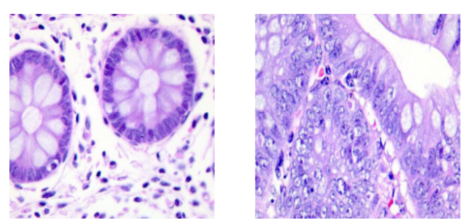

Fig. 2: Sample Histopathological for two colon cancer types)

## IV. Algorithms used

A popular deep learning model in computer vision tasks such as image recognition and classification prior to vision transfomers, is the CNN. CNN's algorithm mimics how the brain interprets visual data by drawing inspiration from the human visual system. Convolutional layers, activation layers, pooling layers, and fully linked layers are some of the layers that make up this system [25], [26]. The first layer of the method is the convolutional layer, which is applied to the input image [27]. The output of the model is frequently transformed into probability scores, which represent the likelihood of each class label, in the last layer using a softmax activation function. The use of CNNs for colon and lung cancer detection has improved computer-aided diagnostic (CAD) tools and stimulated further study and development in the area of medical image analysis [26], [28].

### A. *VGG19*

The Visual Geometry Group (VGG) at the University of Oxford has developed a deep convolutional neural network architecture called VGG19. It is known for its efficiency and simplicity in image identification tasks and is an expansion of the original VGG16 model with 19 layers [29], [30]. With 19 connection layers, comprising 16 convolutional layers and 3 fully connected layers, the VGG-19 is a deep learning neural network. The convolutional layers in the VGG19 method are followed by max-pooling [31] layers for downsampling. The depth of the network, which contains modest 3x3 filters on each convolutional layer, enables it to learn hierarchical features from input photos. The main feature of the architecture is its homogeneity (using only 3x3 filters in convolutional layers), which simplifies the model and makes it simple to use. VGG19 has developed into a well-known benchmark model in the field of deep learning due to its basic architecture and good performance. It also serves as a base for creating more intricate convolutional neural networks. Its consistency of performance and capacity to learn discriminative features have made it a dependable option for a variety of computer vision applications, including medical imaging. VGG19 can be easily modified to perform additional medical imaging tasks, such as segmentation and localization, despite its original design being for image classification. Due to its adaptability, VGG19 can be used by researchers and clinicians to construct thorough cancer detection systems.

### B. *Bottleneck Resnet50*

The Bottleneck ResNet50 is a variant of the Residual Neural Network (ResNet) architecture, which was introduced to address the challenges of training very deep neural networks. Use of a "bottleneck" design in the residual blocks is the primary principle of the Bottleneck ResNet, which aims to reduce computing complexity while preserving the benefits of ResNet. ResNet50 is a specific variant of the Residual Neural Network (ResNet) architecture, where the number "50" indicates the depth of the network [32], [33]. The ResNet50 network has 50 layers, which makes it a very deep model. There is a residual block included in every layer that enables the model to learn residual mappings that may be used on the output of prior layers. This skip connection in the residual block makes it easier to train deep networks and addresses the vanishing gradient issue. A 1x1 convolutional layer is used to reduce dimensionality, a 3x3 convolutional layer is the primary convolutional layer for feature extraction, and a final 1x1 convolutional layer is used to increase dimensionality in the Bottleneck ResNet model's algorithm [34][9]. With less computational complexity due to the bottleneck design in the residual blocks, ResNet50 is easier to train than extremely deep networks without compromising performance. This helps avoid overfitting and enhances generalization, which is especially helpful when working with small medical imaging datasets.

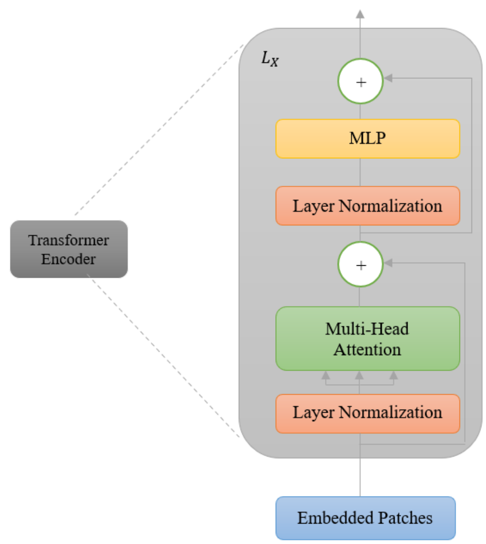


Fig. 3: Encoder Architecture of the Transformer

### C. *Vision Transformer*

The Vision Transformer (ViT) is a novel deep learning architecture that has shown exceptional performance on various visual tasks, especially in the domain of natural images. ViT relies on a self-attention mechanism that enables capturing long-range dependencies and relationships between different image regions. The model divides an input image into fixed-size non-overlapping patches and linearly embeds each patch before feeding them into a transformer encoder. The resulting representations are then utilized for classification tasks. The primary advantage of ViT is its ability to learn meaningful representations from large-scale datasets without the need for hand-crafted features. In contrast to the original transformer type that comprises both an encoder and a decoder, the ViT model solely incorporates an encoder in its architecture [35]. The input image for ViT is represented in the $\Re^{H \times W \times C}$ format and subsequently divided into N patches of size $P \times P$. Here, $P \times P$ is determined by the dimensions H, W, and the number of channels C, specifically $P \times P = \frac{HW}{P^2}$ , where C = number of channels, H = height, and W = width.

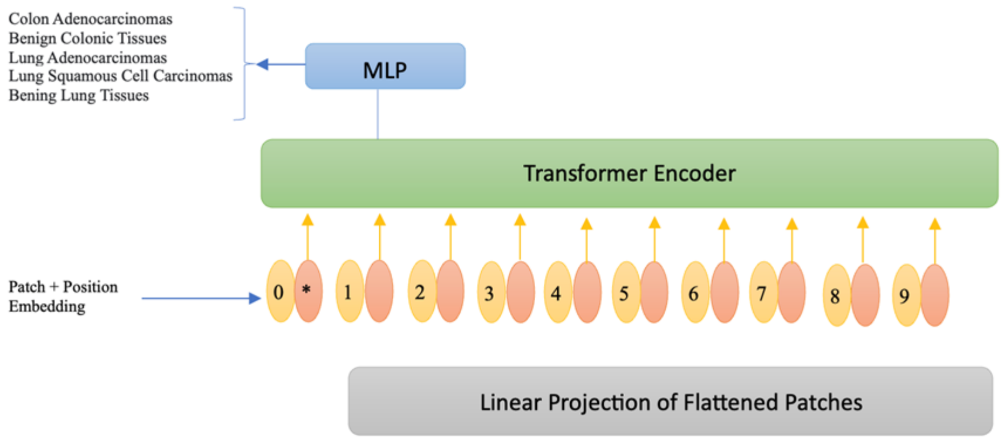


Fig. 4: The Vision Transformer Architecture

For final classification, a learnable patch embedding is thoughtfully incorporated through an MLP head. Subsequently, these combined patches and position embeddings embark on a transformative journey within the transformer encoder model, adorned with alternating layers of multi-headed self-attention and MLP blocks. As clearly shown in Figure 3, the layer normalization (LN) was applied before every block, while residual connections were applied after every block.

Like BERT's [36] approach, a learnable embedding is combined with the sequence of patch embeddings ($Z_0 = I_{class}$). The working principle of ViT is captured through Equations (1)-(4). Equation (1) reveals the essence of positional embedding $E_{pos}$, a matrix of learnable parameters. The embedding of patch N, $X\frac{N}{p}E$, takes form through a learnable linear projection, leading to the remarkable output $z_0$. By integrating position embeddings, a meaningful order is established within the input image patches, amplifying the model's process. In the transformer encoder layer, the first block commences with layer normalization (LN), leading into multi-head self-attention (MSA), yielding the output $\frac{Z'}{l}$ at layer $l$. The second block unfolds, starting with an LN layer, journeying through an MLP with the Gaussian error linear unit (GELU) nonlinearity, and a residual connection, giving us with the output $Z_l$, (described in Equations 2 and 3). This produces a transformer encoder as depicted in Figure 3.

$$Z_0 = \left[I_{class}; X\tfrac{1}{p}E; ...; X\tfrac{N}{p}E\right] + E_{pos}E\epsilon\Re^{(P^2C)\times D}, E_{pos}\epsilon\Re^{(N+1\times D)} \tag{1}$$

$$Z'_l = MSA(LN(Z_{l-1})) + Z_{l-1} \quad l = 1...L \tag{2}$$

$$Z_l = MLP(LN(Z'_l)) + Z'_l \quad l = 1...L \tag{3}$$

$$y = LN(Z_L^{\circ}) \tag{4}$$

## V. Experiment and Results

Four different state-of-art deep-learning models, namely, Convolutional Neural Network (CNN), VGG19, Bottleneck ResNet50, and Vision Transformer (ViT) were used to complete the goal of detecting lung and colon cancer in images. The dataset of lung and colon cancer has five classes of cancers that consist of 25,000 histopathology images. Five classes to detect cancer are (i) Lung benign tissue, (ii) Lung adenocarcinoma, (iii) Lung squamous cell carcinoma, (iv) Colon adenocarcinoma, and (v) Colon benign tissue. The images were subjected to preprocessing to ensure uniform dimensions, appropriate normalization, and augmentation. The dataset was then split into training, validation, and test sets with a distribution of 50%, 25%, and 25% respectively. Each model was constructed using the training set, and the hyperparameters were fine-tuned using the validation set. Finally, the performance of the models was assessed using the test set.

The experimental outcomes revealed diverse performance levels [37] among the four models in lung and colon cancer detection. While VGG19 achieved an accuracy of only 91%, both CNN and ResNet50 models displayed competitive accuracy, delivering results with 99% precision. ViT exhibited enhanced sensitivity compared to both CNN and ResNet50, showcasing its 100% effectiveness in identifying cancerous samples. The confusion matrices for the four models, namely Convolutional Neural Network (CNN), VGG19, Bottleneck ResNet50, and Vision Transformer (ViT), are displayed in

| | CNN | | | VGG19 | | | Bottleneck ResNet50 | | | ViT Transformer | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | precision | recall | f1-score | precision | recall | f1-score | precision | recall | f1-score | precision | recall | f1-score |
| **colon aca** | 1 | 0.9792 | 0.9895 | 0.9241 | 0.8971 | 0.9104 | 0.9992 | 0.9984 | 0.9988 | 1 | 1 | 1 |
| **colon n** | 0.9796 | 1 | 0.9897 | 0.8999 | 0.9839 | 0.94 | 0.9984 | 0.9992 | 0.9988 | 1 | 1 | 1 |
| **lung_aca** | 0.9968 | 0.9952 | 0.996 | 0.81 | 0.9075 | 0.856 | 0.9904 | 0.9928 | 0.9916 | 1 | 1 | 1 |
| **lung_n** | 1 | 1 | 1 | 0.9944 | 0.9936 | 0.994 | 1 | 1 | 1 | 1 | 1 | 1 |
| **lung_scc** | 0.9952 | 0.9968 | 0.996 | 0.9581 | 0.78 | 0.86 | 0.9928 | 0.9904 | 0.9916 | 1 | 1 | 1 |
| **accuracy** | | | 0.9942 | | | 0.9128 | | | 0.9962 | | | 1 |
| **macro avg** | 0.9943 | 0.9942 | 0.9942 | 0.9173 | 0.9124 | 0.9121 | 0.9962 | 0.9962 | 0.9962 | 1 | 1 | 1 |
| **weighted avg** | 0.9943 | 0.9942 | 0.9942 | 0.9173 | 0.9128 | 0.9123 | 0.9962 | 0.9962 | 0.9962 | 1 | 1 | 1 |

TABLE I: CNN

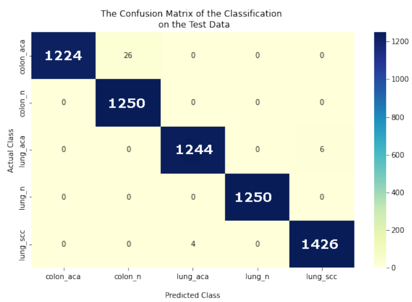


Fig. 5: Confusion Matrix of Bottleneck Resnet50 on the Test Data

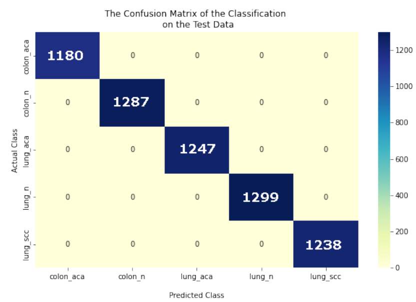


Fig. 7: Confusion Matrix of ViT on the Test Data

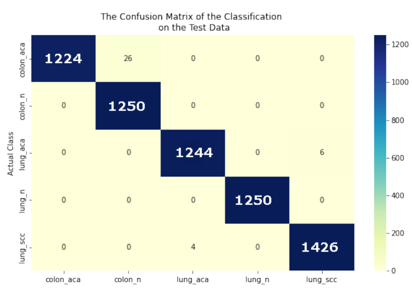


Fig. 6: Confusion Matrix of CNN on the Test Data

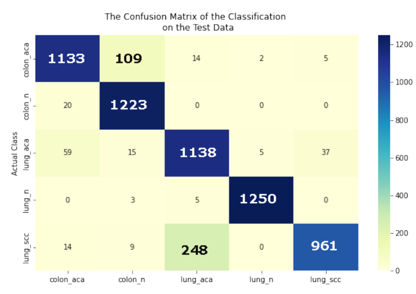


Fig. 8: Confusion Matrix of VGG19 on the Test Data

figures 1, 2, 3, and 4, respectively. Each figure provides insights into the accuracy, precision, recall (sensitivity), and specificity of the corresponding model. These findings indicate that ViT surpasses all other models for lung and colon cancer detection. However, the selection of the most appropriate model may hinge on specific [38], such as the desired balance between sensitivity and specificity, available computational resources, interpretability, and other pertinent factors.

The LC25000 dataset comprises 25,000 histopathology images divided into five classes, with two classes pertaining to colon tissue and three classes associated with lung tissues. For training, 50% of the dataset (12,500 images) was utilized, while the remaining 25% served as the test and 25% served as the validation set. The input image resolution was set to 224 × 224, and through meticulous optimization of hyperparameters, a learning rate of 2e-5 (0.00002) was determined. The model showcased exceptional performance at this resolution, boasting a validation accuracy of 100% on the fifth epoch.

On the test set, the ViT models demonstrated their prowess with 224x224 resolutions, exhibiting a test loss of 0.012622 and an impeccable test accuracy of 100%. The performance was further visualized through informative confusion matrices in Figure 4. The performance on the testing dataset also showed that the ResNet model in Figure 5 performed better than VGG19 8. There was significantly less number of false positives as can be seen in the confusion matrix. This resulted in higher precision and recall values for the ResNet model in both uninfected and parasitized cells as shown in Table I.

## VI. Conclusion

Recently, there has been a lot of promise in the automated identification of colon and lung cancer from histopathology pictures using ViT and deep learning approaches. The research examined in this literature review shows that these methods for enhancing the precision and efficacy of cancer diagnosis are both feasible and effective. Apart from achieving outstanding output with Vision Transformation in the future, we can enhance this research to achieve more accuracy in the deep learning CNN, VGG10, and Resnet 50 algorithms with state-of-the-art preprocessing mechanisms. Further research is needed to validate these approaches on larger datasets and to integrate them into clinical practice by dry running with real-world applications.

## REFERENCES


[1] Cancer — who.int. https://www.who.int/news-room/fact-sheets/detail/cancer. [Accessed 01-08-2023].

[2] J Ferlay, M Ervik, F Lam, M Colombet, L Mery, M Piñeros, A Znaor, I Soerjomataram, and F Bray. Global cancer observatory: cancer today. lyon: International agency for research on cancer; 2020. *Cancer Tomorrow*, 2021.

[3] Takaharu Kato, Koichi Suzuki, Yuta Muto, Junichi Sasaki, Shingo Tsujinaka, Yutaka J Kawamura, Hiroshi Noda, Hisanaga Horie, Fumio Konishi, and Toshiki Rikiyama. Multiple primary malignancies involving primary sporadic colorectal cancer in japan: incidence of gastric cancer with colorectal cancer patients may be higher than previously recognized. *World Journal of Surgical Oncology*, 13:1–7, 2015.

[4] Koich Kurishima, Kunihiko Miyazaki, Hiroko Watanabe, Toshihiro Shiozawa, Hiroichi Ishikawa, Hiroaki Satoh, and Nobuyuki Hizawa. Lung cancer patients with synchronous colon cancer. *Molecular and clinical oncology*, 8(1):137–140, 2018.

[5] Sabrina Islam, Rahagir Salekeen, and Ayesha Ashraf. Computational screening of natural mtbdxr inhibitors for novel anti-tuberculosis compound discovery. *Journal of Biomolecular Structure and Dynamics*, pages 1–11, 2023.

[6] Detection of Diseases Using Machine Learning Image Recognition Technology in Artificial Intelligence — hindawi.com. https://www.hindawi.com/journals/cin/2022/5658641/. [Accessed 01-08-2023].

[7] Mostofa Ahsan, Rahul Gomes, and Anne Denton. Application of a convolutional neural network using transfer learning for tuberculosis detection. In *2019 IEEE International Conference on Electro Information Technology (EIT)*, pages 427–433. IEEE, 2019.

[8] Gong Cheng, Junwei Han, and Xiaoqiang Lu. Remote sensing image scene classification: Benchmark and state of the art. *Proceedings of the IEEE*, 105(10):1865–1883, 2017.

[9] Maryam Mehmood, Ahsan Shahzad, Bushra Zafar, Amsa Shabbir, and Nouman Ali. Remote sensing image classification: A comprehensive review and applications. *Mathematical Problems in Engineering*, 2022:1–24, 2022.

[10] Mostofa Ahsan, Nafiz Rifat, Md Chowdhury, and Rahul Gomes. Detecting cyber attacks: A reinforcement learning based intrusion detection system. In *2022 IEEE International Conference on Electro Information Technology (eIT)*, pages 461–466. IEEE, 2022.

[11] Nafiz Rifat, Mostofa Ahsan, Md Chowdhury, and Rahul Gomes. Bert against social engineering attack: Phishing text detection. In *2022 IEEE International Conference on Electro Information Technology (eIT)*, pages 1–6. IEEE, 2022.

[12] Md Saifur Rahman, Nafiz Rifat, Mostofa Ahsan, Sabrina Islam, Md Chowdhury, and Rahul Gomes. Deep learning application for detection of malaria. In *2023 IEEE International Conference on Electro Information Technology (eIT)*, pages 1–5. IEEE, 2023.

[13] Nafiz Imtiaz Rifat. Feature engineering on the cybersecurity dataset for deployment on software defined network. 2020.

[14] Nafiz Rifat, Mostofa Ahsan, Rahul Gomes, and Md Chowdhury. Covid-19 sentiment analysis applying bert. In *2022 IEEE International Conference on Electro Information Technology (eIT)*, pages 417–422. IEEE, 2022.

[15] Rahul Gomes, Papia Rozario, and Nishan Adhikari. Deep learning optimization in remote sensing image segmentation using dilated convolutions and shufflenet. In *2021 IEEE International Conference on Electro Information Technology (EIT)*, pages 244–249. IEEE, 2021.

[16] Rahul Gomes, Connor Kamrowski, Jordan Langlois, Papia Rozario, Ian Dircks, Keegan Grottodden, Matthew Martinez, Wei Zhong Tee, Kyle Sargeant, Corbin LaFleur, et al. A comprehensive review of machine learning used to combat covid-19. *Diagnostics*, 12(8):1853, 2022.

[17] Yifei Chen, Yi Li, Rajiv Narayan, Aravind Subramanian, and Xiaohui Xie. Gene expression inference with deep learning. *Bioinformatics*, 32(12):1832–1839, 2016.

[18] Redirecting — doi.org. https://doi.org/10.1016/j.bspc.2022.103596. [Accessed 31-07-2023].

[19] Supriya Suresh and Subaji Mohan. Roi-based feature learning for efficient true positive prediction using convolutional neural network for lung cancer diagnosis. *Neural Computing and Applications*, 32(20):15989–16009, 2020.

[20] Shidan Wang, Donghan M Yang, Ruichen Rong, Xiaowei Zhan, Junya Fujimoto, Hongyu Liu, John Minna, Ignacio Ivan Wistuba, Yang Xie, and Guanghua Xiao. Artificial intelligence in lung cancer pathology image analysis. *Cancers*, 11(11):1673, 2019.

[21] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929*, 2020.

[22] Jiangyun Li, Wenxuan Wang, Chen Chen, Tianxiang Zhang, Sen Zha, Jing Wang, and Hong Yu. Transbtsv2: Towards better and more efficient volumetric segmentation of medical images. *arXiv preprint arXiv:2201.12785*, 2022.

[23] Andrew A Borkowski, Marilyn M Bui, L Brannon Thomas, Catherine P Wilson, Lauren A DeLand, and Stephen M Mastorides. Lung and colon cancer histopathological image dataset (lc25000). *arXiv preprint arXiv:1912.12142*, 2019.

[24] Lung and Colon Cancer Histopathological Images — kaggle.com. https://www.kaggle.com/datasets/andrewmvd/lung-and-colon-cancer-histopathological-images. [Accessed 31-07-2023].

[25] Teja Kattenborn, Jens Leitloff, Felix Schiefer, and Stefan Hinz. Review on convolutional neural networks (cnn) in vegetation remote sensing. *ISPRS journal of photogrammetry and remote sensing*, 173:24–49, 2021.

[26] Sanidhya Mangal, Aanchal Chaurasia, and Ayush Khajanchi. Convolution neural networks for diagnosing colon and lung cancer histopathological images. *arXiv preprint arXiv:2009.03878*, 2020.

[27] Mohammed Al-Jabbar, Mohammed Alshahrani, Ebrahim Mohammed Senan, and Ibrahim Abdulrab Ahmed. Histopathological analysis for detecting lung and colon cancer malignancies using hybrid systems with fused features. *Bioengineering*, 10(3):383, 2023.

[28] Radical Rakhman Wahid, Chilyatun Nisa, Rahayu Prabawati Amaliyah, and Eva Yulia Puspaningrum. Lung and colon cancer detection with convolutional neural networks on histopathological images. In *AIP Conference Proceedings*, volume 2654. AIP Publishing, 2023.

[29] Thanh-Hai Nguyen, Thanh-Nghia Nguyen, and Ba-Viet Ngo. A vgg-19 model with transfer learning and image segmentation for classification of tomato leaf disease. *AgriEngineering*, 4(4):871–887, 2022.

[30] Abdulkadir Karacı. Vggcov19-net: automatic detection of covid-19 cases from x-ray images using modified vgg19 cnn architecture and yolo algorithm. *Neural Computing and Applications*, 34(10):8253–8274, 2022.

[31] Sunil GC, Borhan Saidul Md, Yu Zhang, Demetris Reed, Mostofa Ahsan, Eric Berg, and Xin Sun. Using deep learning neural network in artificial intelligence technology to classify beef cuts. *Frontiers in Sensors*, 2:654357, 2021.

[32] Yinghui Liu, Hua Xu, Zisen Qi, and Yunhao Shi. Specific emitter identification against unreliable features interference based on time-series classification network structure. *IEEE Access*, 8:200194–200208, 2020.

[33] Lewis Campbell Boyd, Vanja Popovic, and Jan Paul Siebert. Deep reinforcement learning control of hand-eye coordination with a software retina. In *2020 International Joint Conference on Neural Networks (IJCNN)*, pages 1–8. IEEE, 2020.

[34] Divakar Verma, Manish Kumar, and Srinivas Eregala. Deep demosaicing using resnet-bottleneck architecture. In *International Conference on Computer Vision and Image Processing*, pages 170–179. Springer, 2019.

[35] Sudhakar Tummala, Seifedine Kadry, Syed Ahmad Chan Bukhari, and Hafiz Tayyab Rauf. Classification of brain tumor from magnetic resonance imaging using vision transformers ensembling. *Current Oncology*, 29(10):7498–7511, 2022.

[36] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805*, 2018.

[37] Kendall E Nygard, Mostofa Ahsan, Aakanksha Rastogi, and Rashmi Satyal. Methodological choices in machine learning applications. *International Journal on Advances in Software Volume 16, Number 1 & 2, 2023*, 2023.

[38] Kendall E Nygard, Mostofa Ahsan, Aakanksha Rastogi, and Rashmi Satyal. Data and feature engineering challenges in machine learning.